DEPI Final Project - Database Design & Data Modeling

AI Portfolio Generator Project Documentation

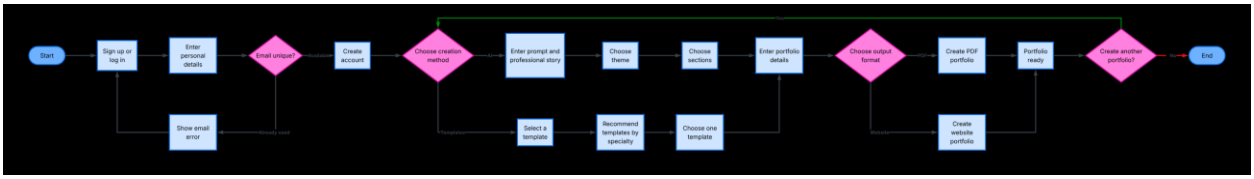
1. Project Overview & Architecture

This document contains the Logical and Physical Schema for the AI Portfolio Generator project developed as part of the DEPI (Digital Egypt Pioneers Initiative) final project internship. The application supports two primary creation pathways for users:

- **Ready Templates Path:** Users select a pre-defined template based on their specialty and fill out structured data (Location, About Me, Design, Projects, Experience, Skills, Social Links).
- **AI Generation Path:** Users enter a prompt and professional story, choose a theme and sections, and the AI generates the portfolio accordingly.

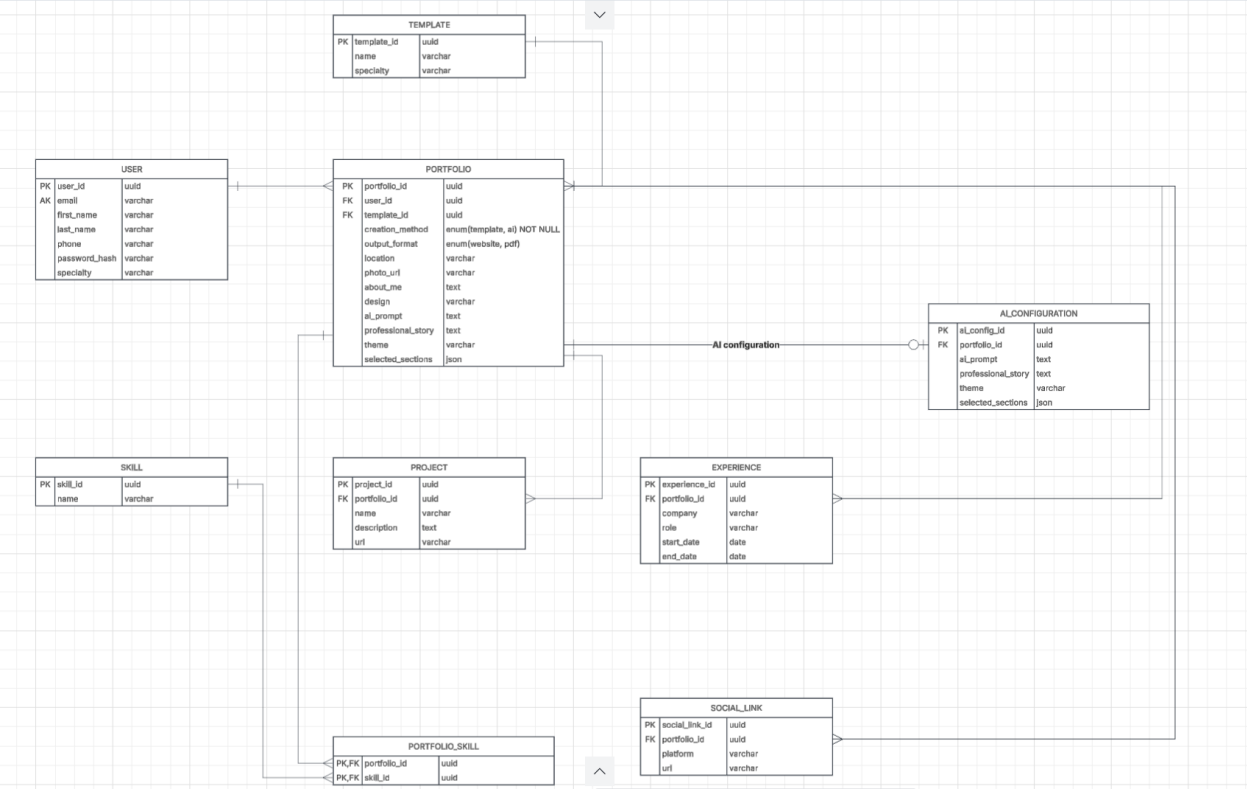
2. System Diagrams (Process Flow & ERD)

The diagrams below illustrate the system process flow and the Entity-Relationship Diagram (ERD) designed for the platform in high resolution:



3. Physical Schema Tables

The tables below outline the physical database schema, data types, and constraints implemented in the database:



Users Table

Column Name	Data Type	Constraints / Description
user_id	VARCHAR(36)	Primary Key (PK)
email	VARCHAR(255)	Unique, Not Null
first_name	VARCHAR(100)	Not Null
last_name	VARCHAR(100)	Not Null
phone	VARCHAR(50)	Optional
password_hash	VARCHAR(255)	Not Null
specialty	VARCHAR(100)	User's field/specialty

Templates Table

Column Name	Data Type	Constraints / Description
template_id	VARCHAR(36)	Primary Key (PK)
name	VARCHAR(100)	Template Name
specialty	VARCHAR(100)	Suitable specialty

Portfolios Table

Column Name	Data Type	Constraints / Description

portfolio_id	VARCHAR(36)	Primary Key (PK)
user_id	VARCHAR(36)	Foreign Key (FK) -> References Users
template_id	VARCHAR(36)	Foreign Key (FK) -> References Templates (Nullable)
creation_method	VARCHAR(50)	Not Null (e.g., 'template' or 'ai')
output_format	VARCHAR(50)	Not Null (e.g., 'website' or 'pdf')
location	VARCHAR(100)	For Templates path
photo_url	VARCHAR(255)	For Templates path
about_me	TEXT	For Templates path
design	VARCHAR(100)	For Templates path
ai_prompt	TEXT	For AI path
professional_story	TEXT	For AI path
theme	VARCHAR(100)	For AI path
selected_sections	JSON	For AI path

Projects Table

Column Name	Data Type	Constraints / Description
project_id	VARCHAR(36)	Primary Key (PK)
portfolio_id	VARCHAR(36)	Foreign Key (FK) -> References Portfolios
name	VARCHAR(150)	Project Name
description	TEXT	Project Details
url	VARCHAR(255)	Project Link

Experience Table

Column Name	Data Type	Constraints / Description
experience_id	VARCHAR(36)	Primary Key (PK)
portfolio_id	VARCHAR(36)	Foreign Key (FK) -> References Portfolios
company	VARCHAR(150)	Company Name
role	VARCHAR(150)	Job Title / Role
start_date	DATE	Start Date
end_date	DATE	End Date

Skills Table

Column Name	Data Type	Constraints / Description
skill_id	VARCHAR(36)	Primary Key (PK)
name	VARCHAR(100)	Skill Name

Portfolio Skills Table

Column Name	Data Type	Constraints / Description
portfolio_id	VARCHAR(36)	PK & FK -> References Portfolios
skill_id	VARCHAR(36)	PK & FK -> References Skills

Social Links Table

Column Name	Data Type	Constraints / Description
social_link_id	VARCHAR(36)	Primary Key (PK)
portfolio_id	VARCHAR(36)	Foreign Key (FK) -> References Portfolios
platform	VARCHAR(50)	Platform Name (e.g., LinkedIn, X)
url	VARCHAR(255)	Profile Link